Probability

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Probability Practice: A bag contains 6 red and 9 blue counters. A counter is picked at random. Find P(red) as a fraction, and P(not red).

Answer: P(red) = 6/15 = 2/5. P(not red) = 1 − 2/5 = 3/5.

Probability Basics (15 questions)

1. A fair six-sided dice is rolled once. What is the probability of rolling a 6? Give your answer as a fraction.
2. A spinner has 5 equal sections numbered 1 to 5. What is the probability of landing on an even number?
3. Write the probability 'impossible' as a number. A) 0 B) 0.25 C) 0.5 D) 0.75 E) 1
4. A bag contains 20 balls, 5 of which are green. What is the probability of picking a green ball, as a decimal?
5. Order these words from least likely to most likely: certain, unlikely, impossible, likely, evens.
6. A coin is flipped once. What is the probability of landing on heads, as a percentage?
7. Which word best describes an event with probability 0.9? A) Impossible B) Unlikely C) Evens D) Likely E) Certain
8. A bag contains only white counters. A counter is picked at random. What is the probability it is black?
9. List all the possible outcomes when a spinner with sections 1, 2 and 3 is spun once.
10. A box contains 9 green pens and 1 orange pen. What is the probability of picking an orange pen, as a fraction?
11. Select ALL that apply: which of these could be the probability of an event? A) 1 B) −0.5 C) 0.75 D) 2 E) 0
12. A fair six-sided dice is rolled once. What is the probability of rolling a number less than 3? A) 1/6 B) 1/3 C) 1/2 D) 2/3 E) 5/6
13. The probability that a bus is on time is 17/20. Write this probability as a percentage.
14. List all the possible outcomes when a coin is flipped and then a second coin is flipped.
15. A spinner is divided into 5 equal sections coloured red, red, blue, green and yellow. What is the probability of landing on red?

Applying Probability (15 questions)

1. A bag contains 6 blue, 2 red and 2 yellow counters. A counter is picked at random. What is the probability it is NOT blue? Give your answer as a fraction.
2. A spinner has 8 equal sections: 5 are white, 3 are black. What is the probability of landing on black, as a decimal?
3. The probability that a football team wins a match is 0.55. What is the probability that they do not win? A) 0.35 B) 0.4 C) 0.45 D) 0.5 E) 0.55
4. A fair coin is flipped twice. List all 4 possible outcomes, then state the probability of getting at least one tail.
5. A jar contains 25 marbles: 10 are striped, the rest are plain. What is the probability of picking a plain marble, as a percentage?
6. Two fair coins are flipped. What is the probability of getting two heads or two tails (not a mix)? A) 1/4 B) 1/2 C) 3/4 D) 1 E) 0

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7. A bag has 5 orange and 3 purple counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is purple, what is the probability the second counter picked is also purple?

8. Select ALL that apply: a fair spinner numbered 1 to 8 is spun once. Which statements are true? A) P(spinning a number greater than 8) = 0. B) P(spinning an odd number) = 1/2. C) P(spinning a number less than 9) = 1. D) P(spinning an 8) = 1/8. E) P(spinning a multiple of 5) = 1/4.

9. A box of 20 biscuits has 12 plain and 8 chocolate biscuits. What is the probability of picking a chocolate biscuit, as a fraction in its simplest form?

10. A weather forecaster says there is an 80% chance of rain tomorrow. What is the probability, as a fraction, that it will NOT rain?

11. A bag contains only green and yellow counters. P(green) = 0.45. If there are 20 counters in total, how many are yellow?

12. A fair spinner has sections numbered 1 to 4. It is spun twice. What is the probability that it lands on 1 both times? A) 1/16 B) 1/8 C) 1/4 D) 1/2 E) 2/16

13. A drawer contains 5 red gloves and 5 grey gloves. One glove is pulled out at random, not put back, and a second glove is pulled out. What is the probability that both gloves are red?

14. In a raffle, 150 tickets are sold and Tom buys 3 of them. What is the probability that Tom's ticket is drawn as the winner, as a decimal?

15. Select ALL that apply: a bag contains 2 red, 5 blue and 3 green counters (10 total). Which statements are true? A) P(red) = 1/5. B) P(not blue) = 1/2. C) P(green) = 0.3. D) P(blue) = 0.5. E) P(red or green) = 0.6.

Standard / Advanced (15 questions)

1. A bag contains 9 counters, 3 of which are red. If the probability of picking a red counter stays the same, how many red counters would there be in a bag of 36 counters of the same mix?

2. A fair coin is flipped 3 times. How many total possible outcome sequences are there (e.g. HHH, HHT, ...)? A) 4 B) 6 C) 8 D) 10 E) 12

3. A spinner has equal sections numbered 1 to 10. What is the probability of spinning a factor of 12?

4. A bag has 3 red and 1 blue counter. What is the probability of picking the blue counter, expressed as a percentage, then doubled?

5. Two fair dice are rolled and their scores are added together. What is the most likely total, and what is its probability as a fraction?

6. A box contains red, blue and green counters only, in the ratio 1:2:3. A counter is picked at random. What is the probability it is NOT red?

7. A bag has 5 black and 7 white counters. A counter is picked and not replaced, then a second is picked. What is the probability that both counters are white? A) 7/12 B) 35/132 C) 49/144 D) 7/22 E) 7/24

8. A weather app shows a 25% chance of rain on Saturday and, independently, a 40% chance of rain on Sunday. What is the probability that it rains on both days?

9. A bag contains only red and blue counters. If P(red) = 2/7 and there are 20 blue counters, how many counters are in the bag in total? A) 28 B) 30 C) 32 D) 35 E) 36

10. In a group of 50 students, the probability a randomly chosen student studies French is 0.6. Of those who study French, 1/3 also study Spanish. What is the probability that a randomly chosen student studies both French and Spanish?

11. A spinner numbered 1 to 6 is spun twice and the two results are added. What is the probability that the total is an odd number?

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12. Select ALL that apply: a bag contains 6 red and 4 blue counters. Two counters are picked one after another without replacement. Which statements are true? A) P(first is red) = 0.6. B) P(both red) = 1/3. C) P(first red, second blue) = 4/15. D) P(both blue) = 2/15. E) P(both red) = P(both blue).

13. A fair six-sided dice is rolled twice. What is the probability that the product of the two scores is greater than 20?

14. A bag contains 12 counters: some red, some blue. A counter is picked, replaced, and this is repeated many times; red is picked 3 times as often as blue on average. How many of the 12 counters are red?

15. A box contains 10 tickets numbered 1 to 10. Two tickets are drawn together at random (order does not matter). What is the probability that both tickets drawn are even numbers?

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